

MIE245 - Winter 2025

Lab 9 - Prim's Algorithm

Deadline: Tuesday, March 24, 11:59pm EST

Submission Instructions

Reminder: You can not claim work that was completed by others or generative AI as your own. Submit a completed **readme** template file outlining what code you contributed and what code was completed by other students, people, tools, and/or leveraged from websites.

Add `_utorid` (with `utorid` replaced by your UTORid in all lower-case characters) to the end of both the completed template file names (i.e., the `.py` and the `readme` files). Please ensure you use your UTORid (i.e., what you use to sign into Quercus and **not** other identifiers such as your student number).

Compress your completed template files to a zip file and upload the zip file to Quercus.

Lab Instructions

In this lab, you will write a implementation of Prim's Algorithm (**Eager**).

Complete the template file **MIE245_Lab_9_Template.py** by finishing the following required methods:

- *prims_eager*(*G*, *r*): performs Prim's Algorithm (eager) to identify the minimum spanning tree (MST). Takes a Graph object *G* and source node *r* and updates the vertices' π and key values in *G*'s adjacency list to determine the MST, then returns *G*.

AutoGrading

Additionally, you have access to an example of the auto-grading script that will be used to test your submission (**MIE245_Lab_9_Autograder.py**). Please follow the steps below to run the auto-grader for the provided test case:

1. Place the auto-grader script in a new folder
2. Inside the newly created folder, create a folder called "submissions"
3. Place your script in the "submissions" folder
4. Run the auto-grader script and open the **grades.csv** file to see if you passed the test case (1 is pass, 0 is fail)

Note: The auto-grading script only contains one of the several test cases we will evaluate against, so even if you pass this test case, you may not have a completely correct solution.

Tip: You can modify the test case to test additional functionality.